

Detailed and Updated Summary:
*A review of three-dimensional vision techniques in food and
agriculture applications*
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Based on the original PDF + some clarifications

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Abstract

This systematic review introduces and analyzes various 3D vision techniques employed in food and agricultural systems, often enhanced by deep learning. The objective is to provide a comprehensive overview of operating principles, analysis methods, and applications, helping researchers select the most appropriate 3D imaging system for specific contexts. The examined techniques include structured lighting-based imaging, multi-view systems (Stereo and SfM), Time of Flight (ToF), LiDAR, and depth estimation from monocular images.

1 Introduction and Context

Computer vision is a fundamental component of smart food and agricultural systems, especially due to recent advancements in artificial intelligence and deep learning techniques.

Historically, Red-Green-Blue (RGB) color imaging systems have been common, providing information on color, texture, and two-dimensional (2D) morphology. However, 2D systems fail to record the crucial **3D spatial position** information of objects, which is essential for describing physical properties such as dimensions and for guiding autonomous robotic operations (such as pruning, harvesting, and cutting). Widespread adoption of 3D techniques is expected in the future, facilitated by the availability of low-cost hardware.

1.1 Objectives of the Review

While previous reviews have focused on generic computer vision systems or artificial intelligence models, this is the first study to specifically summarize the **3D imaging techniques** used in the sector.

2 3D Image Acquisition and Reconstruction Systems

3D acquisition systems rely on different physical principles to determine spatial depth.

2.1 Structured Lighting-Based 3D Imaging System

The fundamental principle of these systems is **triangulation**.

2.1.1 Operating Principle (Triangulation)

The system is generally composed of an illuminant (for example, a laser or a projector) and a regular 2D camera. By projecting structured light onto the surface, we can compute the shape of the object through the difference between the projected pattern and the perceived one using the equation:

$$h = \frac{dL}{d + B} \quad (\text{Triangulation Scheme})$$

- **h (Height / Depth):**
 - *Description:* It is the **3D measurement** to be determined. It represents the height of the object's point relative to the **reference plane** (often defined at a distance L).
 - *Unit:* Unit of length (e.g., meters, millimeters).
- **d (Disparity):**
 - *Description:* It is the **2D measurement** observed and calculated. It is the displacement (or distance) on the camera sensor surface between the position of the projected point on the object and the position of the projected point on the reference plane. This variation is exclusively due to the height h of the object.
 - *Role:* It is the independent variable measured directly by the camera that determines depth.

- **L (Distance to Reference Plane):**

- *Description:* It is the fixed, pre-calibrated, and known distance between the camera and the **reference plane** (the plane where the object is assumed to be if it had no height, or the surface of the conveyor belt, thus where the object rests and not its closest surface).
- *Role:* It acts as a scaling factor in the formula.

- **B (Baseline):**

- *Description:* It is the fixed and known physical distance between the projection center of the **illuminant** (laser or projector) and the optical center of the **camera**.
- *Role:* It determines the sensitivity of the system. A wider baseline (larger B) increases the measurement accuracy of disparity but can also increase occlusions.

2.1.2 Selection of Light Sources

Active sources, used to provide trackable points for triangulation, are classified into two categories:

1. Laser-Based Lighting Sources:

- *Dot Lasers:* Often require rotating mirror devices or DMDs (Digital Micromirror Devices) for two-dimensional XY scanning. The scanning speed must match the camera acquisition speed, which can be impractical (MHz).
The laser at a known position casts a laser dot onto the object and the camera acquires it; with two rotating mirrors, the position where the dot falls is changed and the acquisition is performed again.
- *Line Lasers:* Reduce scanning time compared to dot lasers. They can be:
 - **Passive:** The object moves (e.g., via a conveyor belt) through the field of view of the fixed laser. Suitable for high-throughput inspection.
 - **Active:** A rotating mirror reflects the laser line, scanning static objects. It preserves high resolution, but depth resolution can vary due to the angle of incidence.

Advantages: Generally offer a **high depth reconstruction resolution**.

Disadvantages: Low reconstruction speed; need for extensive calibration and synchronization. Obstruction can cause loss of depth information.

2. Projector-Based Lighting Sources:

- Project a series of **distinct patterns** (single coded pattern or multiple binary/grayscale patterns).

Advantages: Avoid slow laser scanning and can achieve **real-time depth map reconstruction**. Less sensitive to light interference compared to color-coded patterns.

Disadvantages: Hardware/software requirements to spatially differentiate patterns limit spatial resolution and localization accuracy. Easily influenced by external environmental conditions.

2.1.3 Applications (SL)

They are the primary choice in indoor food production (Structured Light), where lighting is controlled. Examples include estimating the weight of food products (herrings, bread, muffins) with high accuracy ($R^2 = 0.98$) and measuring quality traits in eggs (yolk index, air cell height). They are also used in plant phenotyping (measuring leaf length and width).

2.2 Multi-View Imaging System

2.2.1 Stereo Vision

Stereo vision is the simplest category of multi-view systems and relies on the **position disparity** of objects in images taken from two different angles (binocular). It is a cost-effective and efficient solution.

The geometric relationship for a point in 3D space, assuming aligned cameras (it is not the camera that moves but the object), is given by:

$$z = \frac{f \cdot B}{x_l - x_r} \quad (\text{Depth Equation (1)})$$

Where:

- f is the focal length of the camera (distance between the acquisition sensor and the theoretical focal point).
- z is the height of the measured point.
- B is the baseline distance between the two cameras.
- x_l and x_r are the coordinates of the projected point on the left and right camera image, respectively.

Stereo Matching Algorithms The key step is finding corresponding points in the left and right images to generate disparity maps. After rectification (a mathematical transformation that virtually aligns the two image planes), the search is constrained to epipolar lines.

Disclaimer: Epipolar geometry is the intrinsic geometric relationship between two cameras looking at the same scene. This geometry dictates that if the camera has been calibrated, a 3D point in the world (and its projection $\mathbf{P}_l$ in the left image) must project onto the right image at a point $\mathbf{P}_r$ that necessarily lies along a specific line called the **epipolar line**.

- **Traditional Methods:**

- *Local Methods*: Low computational cost but low reconstruction quality. The goal is to find the block with the lowest cost (minimum intensity difference, for example) using cost functions such as:
 - * SAD (Sum of Absolute Differences): $\text{SAD}(x, y, d) = \sum_{x, y \in W} |I_l(x, y) - I_r(x, y - d)|$.
 - * SSD (Sum of Squared Differences): $\text{SSD}(x, y, d) = \sum_{x, y \in W} (I_l(x, y) - I_r(x, y - d))^2$.
 - * NCC (Normalized Cross-Correlation).
- *Global Methods*: Utilize all global information, high accuracy but slower. They aim to minimize an energy function (e.g., intensity differences) using tools like Graph Cut or Markov Random Field.
- *Semi-Global Methods*: Excellent compromise between local and global methods. The Semi-Global Block Matching (SGBM) algorithm is common and aggregates matching costs in all 1D directions.

- **Deep Learning (DL) Based Stereo Matching:**

- *MC-CNN*: First DL version, uses Siamese networks to identify similarities between blocks, improving similarity metrics.
- *GC-Net (End-to-End Solutions)*: Generate an intermediate cost volume (Height $\times$ Width $\times$ (Max Disparity + 1) $\times$ Feature Size) to map stereo pairs directly to disparity (depth) maps. They offer significantly better accuracy, but 3D convolutional operations are still time-consuming, making it difficult to meet real-time video streaming requirements for high-resolution applications.

Applications (Stereo Vision) Stereo systems are suitable for both indoor and outdoor applications and are often integrated into autonomous robotic systems for visual guidance and navigation (e.g., harvesting apples, grapes, cotton). They are also used in animal production to measure feed intake, estimate weight, and characterize locomotor behavior.

2.2.2 Structure from Motion (SfM)

SfM is a more generic extension of stereo vision, using a series of two or more images from different viewpoints to reconstruct the complete 3D profile of an object.

SfM Pipeline

1. **Correspondence Search:** Feature extraction (e.g., SIFT) and matching to identify similar features across different views.
2. **Incremental Reconstruction:** Starting from two image pairs, corresponding keypoints are gradually reconstructed and added to a 3D point cloud.
3. **Bundle Adjustment:** Optimization of reconstructed points and camera poses to minimize reprojection error.

Applications (SfM) SfM generates a complete and detailed 3D profile, useful for quantification of complex plant traits (maize phenotyping, canopy structure). It can be integrated with UAV (drone) platforms for large-scale topographic monitoring.

2.3 Time of Flight (ToF) and Light Detection and Ranging (LiDAR)

2.3.1 Operating Principle

Both systems measure distance based on the **time delay** or **phase shift** between emitted light and the return echo.

- **ToF (Generic):** Illuminates the scene using a modulated light source.
- **LiDAR:** Uses laser scanning as a light source to quantify distance.

LiDAR Components and Technology A complete LiDAR system includes laser sources, photodetectors, and a scanning system. The most common scanning systems are:

- *Mechanical Rotating:* Popular solution, high signal-to-noise ratio, but bulky instruments sensitive to vibrations.
- *MEMS:* Miniaturization of the mechanical rotating component.
- *Flash:* Uses an optical diffuser, but performance is limited by laser power for long-range measurements.
- *Optical Phased Array:* Modulates the phase of light to avoid time-consuming laser scanning, ideal for high-vibration applications.

Advantages and Applications

- **ToF**: Performance and cost comparable to stereo cameras, but more suitable for **indoor** applications or **low-light** conditions. Used for mass and volume estimation (tomatoes, eggs) and animal behavior tracking (laying hens, cows).
- **LiDAR**: More suitable for **long-range** measurements. Widely adopted for high-throughput phenotyping (measuring crop height, estimating yield and biomass). It is also fundamental for autonomous vehicle navigation and agricultural field mapping (SLAM). UGV (ground vehicle) systems with LiDAR can extract high-resolution vertical profiles, including plant organ-level information.

Disadvantages Generic ToF sensors have low depth reconstruction accuracy and limited spatial resolution. High-resolution LiDAR systems are expensive, and the point cloud density is relatively low for guiding precision robotic operations.

2.4 Depth Estimation from Monocular Image (Single Camera)

This method does not require additional information such as stereo views or optical flow, significantly reducing hardware costs.

This is due to the fact that incident light is reflected in a perfectly diffuse manner in all directions.

- **Traditional Methods (Shape from Shading)**: Model light sources and the object using the Lambertian assumption. The 3D shape is estimated by minimizing prediction error, but it is sensitive to environmental interference and noise. Initial applications include the reconstruction of apple defects.

Lambertian Assumption: An ideal Lambertian surface is a surface that appears equally bright from any viewing angle, provided it is uniformly illuminated. This is due to the fact that incident light is reflected in a perfectly diffuse manner in all directions (it is not the incident brightness that matters but the light intensity).

- **Learning-Based Methods**: Remodel the problem as a continuous pixel-level regression, using neural networks to learn the mapping from raw RGB images to the actual depth (ground truth). An example is MonoDA net, a self-supervised model for 3D scene perception in vineyards.

The main challenge lies in the need for a **vast dataset** for training and the difficulty of **generalizing** models to unseen scenes and environments.

3 3D Image Analysis Methods

Analysis of 3D data is mainly divided into two categories, depending on the sensor's output format.

3.1 RGB-Depth (RGB-D) Image Analysis

The RGB-D data format is typically represented as an $X \times Y \times 4$ image, where the four channels are (R, G, B, Depth).

Analysis Techniques

- **Depth Thresholding:** Simplest idea; uses a depth threshold to filter out the background and interference. This helps remove unwanted objects like leaves or branches for fruit identification.
- *Low-Level Features:* Generation of gradient fields from the depth image to identify vorticity centers (points where the gradient changes direction or vanishes rapidly, useful for fruit detection).
- **Deep Learning-Based Fusion:**
 - *Input-level fusion:* Adding the depth channel to the RGB input (simple method).
 - *Feature extraction-level fusion:* Initially, RGB and depth are processed in two different branches, then the deep features extracted from the two branches are concatenated (often a more robust method).

3.2 Point Cloud Analysis

The point cloud is an **unstructured and unordered** data format, represented as an $N \times 3$ matrix, where N is the number of points and "3" represents the physical 3D coordinates (X, Y, Z). This format best describes the complete 3D profile of the object.

Point Cloud Properties

- **Permutation Invariance:**

Means that the output of an algorithm must not change even if the order (the sequence) of points in the point cloud is modified. Point cloud data are inherently unordered, so processing must be insensitive to the input order.
- **Spatially Varying Density:**

This is a typical **challenge** of point clouds. Due to distance from the sensor, occlusions, or calibration, point density can vary significantly across different areas of the scene. Algorithms must be robust to this variation and not depend on uniform density.
- **Transformation Equivariance:**

Means that the operation on data (for example, feature extraction) must "follow" the transformation applied to the point cloud. If the point cloud is rotated or translated, the internal representation (the features) must rotate or translate correspondingly, keeping the geometric relationship intact.

Initial Processing Phases

- **Registration:** Necessary when data are collected from mobile platforms (UGV, UAV) since LiDAR devices measure distances relative to themselves. The classic algorithm is **Iterative Closest Point (ICP)**, which iteratively minimizes point-to-point or point-to-plane distances between two sets of point clouds.

Once the Point Cloud is obtained, objects must be identified:

Segmentation and Classification Methods

- **Traditional Methods:** Extract information based on geometric constraints of adjacent points, for example, fitting cones and cylinders to segment maize organs.
- **Learning-Based Methods (Deep Learning):** Require reformatting of raw data:
 - **Voxel Representation:** Transforms discrete points into dense 3D cubes, analyzable with 3D convolutional neural networks (3D CNN) (e.g., VoxNet). It is computationally expensive.
 - **Multi-View Representation:** Uses multiple 2D images from different angles to represent the 3D shape, applying CNN to each 2D image (e.g., reprojection onto 3D for segmentation).
 - **PointNet / PointNet++:** Solutions that work **directly** on unordered sets of points, using shared-weight Multi-Layer Perceptrons. PointNet++ improves performance by considering hierarchical local structures. These are used for livestock segmentation.
 - **Graph Neural Networks (GNN):** Another way to encode the point cloud following its unordered characteristic, differing from PointNet because it also considers the features of past updates; furthermore, it structures points by creating a graph node for each of them (particularly effective at modeling local interactions between points).

4 Schematic Concept Map: 3D Imaging Principles

Technique	Typical Cost (Range)	Main Advantages
Structured Lighting (Laser)	~ \$10,000	High depth reconstruction resolution.
Structured Lighting (Projector)	~ \$10,000	High reconstruction speed (for stationary objects).
Stereo Vision	~ \$100 – \$1,000	Low cost; easy customization; video streaming speed.
SfM (Structure from Motion)	~ \$100 – \$1,000	Ability to generate complete 3D profiles.
ToF (Generic)	~ \$100 – \$500	Effective under low light; economical.
LiDAR	~ \$1,000 – \$300,000	Long working distance.
Monocular Depth (DL)	~ \$50 – \$500	Cost-effective and efficient solution.

5 Challenges and Future Opportunities

Despite the growing adoption of 3D vision, significant challenges still exist that open up new research directions.

5.1 Hardware Perspective

- **Cost vs. Performance:** Commercial RGB-D cameras are cheap but have low resolution and are easily influenced by ambient interference (especially outdoors). Industrial laser scanners offer high resolution but are expensive and not always suitable for real-time applications.
- **Future Direction:** It is fundamental to design **low-cost, transferable, reliable, high-resolution, and high-speed** 3D systems.
- **Multi-View Optimization:** Selecting appropriate viewing angles can decrease costs and minimize occlusions.
- **Robotic Integration:** Integrating 3D imaging with robotic platforms is crucial to improving operational capabilities.

5.2 Software Perspective

- **Semantic Depth Understanding:** Differentiating bioproducts with similar textures and depth positions remains very difficult.
- **Lack of Open Data:** There are no open-access 3D datasets available for every single agricultural commodity, hindering the expansion of deep learning in the sector.
- **Multi-Modality Fusion:** Integrating 3D sensors (depth/point cloud) with other modalities (RGB, thermal, hyperspectral, X-ray) can provide more complete information on physicochemical properties.
- **Registration Challenges:** Since different sensing modalities have different resolutions, fields of view, response times, and sampling frequencies, **registration** and the selection of useful features are critical for creating robust fusion models.

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